

CSC311 Introduction to Machine Learning Week 9 Exercises

When submitting problems, please **clearly indicate which question you are submitting**. Use the full question number shown below.

Training

- W9-T1. Consider this news story¹, where when a generative AI tool was asked to make a woman's headshot more 'professional', the tool modified the headshot by changing the woman's race. Explain why a model might behave this way. Is this behaviour intentional? If not, does this absolve the model designers of the impact of this model?
- W9-T2. Explain why a model's overall accuracy is not sufficient to guarantee that a model will "work well" in practice. Why might it be important to consider model performance for sensitive subgroups?
- W9-T3. Explain why, in the temperature prediction problem in lab 2, we did not randomly sample our test set. Why does it make more sense to instead use the final year's data as our test data?
- W9-T4. Suppose that you would like to build a stock market prediction tool to predict the future value of stocks on the TSX. You begin by download all historical stock value data for all of the stocks currently listed on the TSX, with the intention of using this data to train your model. How is this dataset *different* from the data set that your model will encounter at inference time?
- W9-T5. Suppose that we remove a person's sensitive attribute (e.g., gender, race, etc.) from the list of features used to train a classifier. Can we guarantee that this classifier will no longer be biased with respect to these sensitive attributes?
- W9-T6. What is the difference between *disparate treatment* and *disparate impact*?
- W9-T7. Define the following fairness metric. Intuitively, what do each of these metrics consider "fair"?
- (a) demographic parity
 - (b) accuracy parity
 - (c) individual fairness
- W9-T8. Is there a consensus on which of these fairness metrics are "best"? Explain.
- W9-T9. Explain why, in the machine learning challenge, it is inappropriate to split the survey data randomly by response. That is, why is it inappropriate to randomly place the rows of data in training, validation, and test set (e.g., by shuffling the rows, then taking the first X rows for training, the next Y rows for validation, and the remaining rows for the test set?).

Validation

1. What does the term "data leakage" mean? Explain how each of the following are examples of data leakage.
 - (a) Pre-processing on the train-test set
 - (b) Feature selection on the train-test set
 - (c) Temporal leakage
 - (d) Non-independence with the training set

¹<https://www.bostonglobe.com/2023/07/19/business/an-mit-student-asked-ai-make-her-headshot-more-professional-it-gave-her-li>

Generalization

1. In the International AI Safety Report 2025, risks are classified as coming from three main sources: risks from malicious use, risks from malfunction, and systemic risks. Most of the issues we discussed during lecture are risks from malfunction. What is an example of an AI risk from malicious use? How is it different from risk from malfunction?
2. We discussed in class that we can encourage fairness by introducing a fairness regularizer. Consider a binary classification problem. As before, we use $y^{(i)}$ and $t^{(i)}$ to represent the predicted probability and the ground truth class label. Suppose also that some of the data points come from members of a sensitive/protected group, with $s^{(i)} \in \{0, 1\}$ indicating membership in the protective group ($s^{(i)} = 1$ means that data point i comes from the protected group).

Consider the following regularizer

$$\mathcal{R} = \left| \frac{\sum_{i \in S_1} \mathcal{L}_{CE}(y^{(i)}, t^{(i)})}{|S_1|} - \frac{\sum_{i \in S_0} \mathcal{L}_{CE}(y^{(i)}, t^{(i)})}{|S_0|} \right|$$

where:

- $\mathcal{L}_{CE}$ is the cross entropy loss.
 - $S_1 = \{i \mid s^{(i)} = 1\}$ is the set of protected group samples,
 - $S_0 = \{i \mid s^{(i)} = 0\}$ is the set of unprotected group samples,
 - $|S_1|$ and $|S_0|$ denote the number of samples in each group.
- (a) The regularized (cross-entropy) cost can be written $\mathcal{E} = \mathcal{E}_C E + \lambda \mathcal{R}$. Explain what λ represents: what happens if λ is too small? What about too large?
 - (b) Which notion of fairness is used in this regularizer? Demographic parity, accuracy parity, or individual fairness?
 - (c) Convince yourself that $\mathcal{R}$ is differentiable with respect to the predictions (the y s). Why is this important?